

Software Design Document (SDD)

School Management System

Submitted to: Md. Nazmul Hasan

Submitted by: Udipta Mondal

Table of Contents

1. [Introduction]
2. [System Overview]
3. [System Architecture]
4. [Database Tables and Schema]
5. [Module Design]
6. [User Interface Design]
7. [Security Design]
8. [Error Handling]
9. [Performance Considerations]
10. [Testing Strategy]
11. [Deployment Strategy]
12. [Maintenance and Support]
13. [Appendices]

Introduction

Purpose

This Software Design Document (SDD) provides a comprehensive architectural overview of the School Management System. It presents the detailed design for implementing the requirements specified in the project requirements document.

Project Overview

The School Management System is a desktop-based application designed to manage school operations including student admissions, attendance tracking, fee management, teacher management, and class scheduling. The system serves administrators, teachers, and students with appropriate access controls.

Scope

The system will handle:

- Student information management
- Teacher information management
- Class and exam scheduling
- Attendance tracking
- Fee management
- Document uploads and sharing
- User authentication and authorization

Definitions, Acronyms, and Abbreviations

SMS: School Management System

UI: User Interface

DB: Database

MVC: Model-View-Controller

ERD:Entity Relationship Diagram

CRUD:Create, Read, Update, Delete

References

- Project Requirements Document
- System Analysis Document
- C# Programming Standards
- MS SQL Server Documentation

System Overview

System Context

The School Management System operates within a school environment, handling administrative tasks to reduce manual work and improve efficiency. It serves as a central platform for information management and communication between administrators, teachers, and students.

System Functions

The primary functions of the system include:

- Managing student records and admissions
- Managing teacher information
- Tracking attendance
- Managing fees and payments
- Scheduling classes and exams

- Document sharing
- Access control based on user roles

User Classes and Characteristics

The system supports three main user types:

1. Administrators
2. Teachers
3. Students

Operating Environment

Hardware: Desktop/laptop computers with minimum 4GB RAM, 500GB storage

Software: Windows 10/11

Network: Local area network with central server

Database: MS SQL Server 2019 or later

System Architecture

Architectural Overview

The School Management System follows a three-tier architecture:

1. Presentation Layer (UI Layer)
2. Business Logic Layer
3. Data Access Layer

Student Table

```
CREATE TABLE Student (
```

```
StudentID INT PRIMARY KEY,  
Name VARCHAR(100),  
Age INT,  
Gender VARCHAR(10),  
Address VARCHAR(255),  
EnrollmentDate DATE,  
ClassID INT,  
FeesStatus VARCHAR(20),  
TeacherID INT,  
FOREIGN KEY (ClassID) REFERENCES Class(ClassID),  
FOREIGN KEY (TeacherID) REFERENCES Teacher(TeacherID)  
);
```

Teacher Table

```
CREATE TABLE Teacher (  
TeacherID INT PRIMARY KEY,  
Name VARCHAR(100),  
Department VARCHAR(50),  
ContactDetails VARCHAR(255),  
AssignedClassID INT,  
FOREIGN KEY (AssignedClassID) REFERENCES Class(ClassID)  
);
```

Class Table

```
CREATE TABLE Class (  
ClassID INT PRIMARY KEY,  
ClassName VARCHAR(50),
```

```
TeacherID INT,  
  
Schedule VARCHAR(255),  
  
FOREIGN KEY (TeacherID) REFERENCES Teacher(TeacherID)  
  
);
```

Attendance Table

```
CREATE TABLE Attendance (  
  
AttendanceID INT PRIMARY KEY,  
  
StudentID INT,  
  
ClassID INT,  
  
Date DATE,  
  
Status VARCHAR(20),  
  
FOREIGN KEY (StudentID) REFERENCES Student(StudentID),  
  
FOREIGN KEY (ClassID) REFERENCES Class(ClassID)  
  
);
```

Fees Table

```
CREATE TABLE Fees (  
  
FeeID INT PRIMARY KEY,  
  
StudentID INT,  
  
Amount DECIMAL(10,2),  
  
DueDate DATE,  
  
PaidStatus VARCHAR(20),  
  
FOREIGN KEY (StudentID) REFERENCES Student(StudentID)  
  
);
```

ExamSchedule Table

```
CREATE TABLE ExamSchedule (  
    ExamID INT PRIMARY KEY,  
    ClassID INT,  
    ExamDate DATE,  
    Subject VARCHAR(50),  
    Duration INT,  
    FOREIGN KEY (ClassID) REFERENCES Class(ClassID)  
);
```

Module Design

Admin Module

- Student Management
- Teacher Management
- Fee Management
- System Configuration

Functions

-Student Management:

- Add new student
- Update student information

- Search students
- View student details
- Delete student records (with proper authorization)

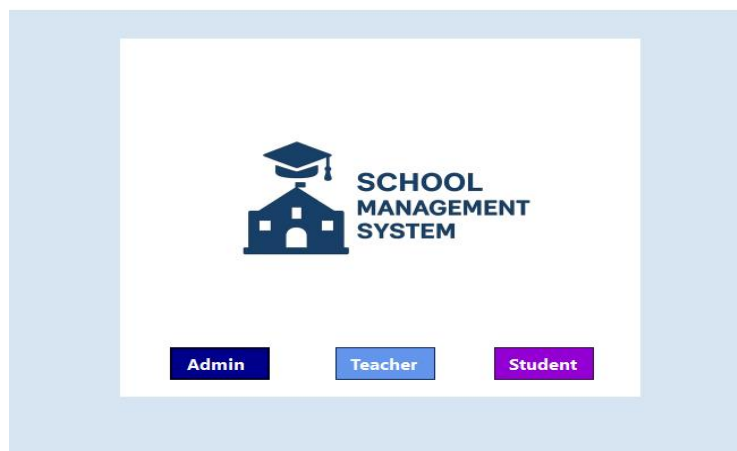
- Teacher Management:

- Add new teacher
- Update teacher information
- Search teachers
- View teacher details
- Assign teachers to classes

- Fee Management:

- Set fee structures
- Record payments
- Generate fee reports
- Send payment reminders

User Interface Design



**Admission**
Admissions

**Fees**

**Student Details**

**Teachers**

**About Us**
About Us

**Exit**

New_Admission



Full Name

Gender

☐ MALE ☐ FEMALE

Dath of Birth

4/ 9/2025

Mobile NO

Father's Name

Mother's Name

Email ID

Address

Submit

Reset

ViewAdmissionA



Admitted Student

	fname	gender	dob	mobile	mname	fname	email	address
▶	dcsc	MALE	2/26/2025	1222	dd	dcd	as@gmail.com	edww
	utsho	FEMALE	2/26/2025	1700891972	ss	asas	sssqsqsq	aasasa
	dipto	MALE	2/28/2025	1700891972	seema mondal	bashudeb mon...	ahondipto@g...	dhaka,Banglad
	dipto	MALE	2/28/2025	1700891972	seema mondal	bashudeb mon...	ahondipto@g...	dhaka,Banglad
	sara	FEMALE	2/13/2025	7899998	sd	sd	sara@gmail.com	dhaka,banglad
	ghgg	MALE	3/19/2025	600977	hj	njyh	nghy	yjyujjyju
	rion	MALE	3/19/2025	10202220	shuli samadder	Ripan sammader	rion@gmail.com	dhaka
	ABc	MALE	3/19/2025	12345	sd	ass	as@gmail.com	sd
	a	MALE	4/4/2025	222	d	ss	ddd	ddd
	rahim	MALE	4/4/2025	1700	ayesha	karim	karim@gmail.c...	dhaka
	rahim	MALE	4/4/2025	12345	karima	karim	rahim@gmail.c...	dhaka



fees

Student id

Full Name

Email Id

Fees

Submit

Search Student

Student ID :

222

Search

	Id	StudentId	FullName	Email	FeeAmount
▶	2	222	Dipto	udipto@gmail.com	20000
•					

←1

i

Basic Information

Full Name

Gender

Date of Birth

Mobile Number

Email

Address

Save

Male

Female

Wednesday,

April

9

▼

Teacher Information

	Id	FullName	Email	Gender	DateOfBirth	MobileNumber	Addre
▶	1	udipta	dipito@gmail.c...	Male	4/8/2025	017	dhaka
	2	Rion	teacehr@gmail...	Male	4/8/2025	213	dhaka
	3	Karim	karim@gmail.c...	Male	4/8/2025	1235	Dhaka
•							

←

→

←1

Class Schedule

Attendance

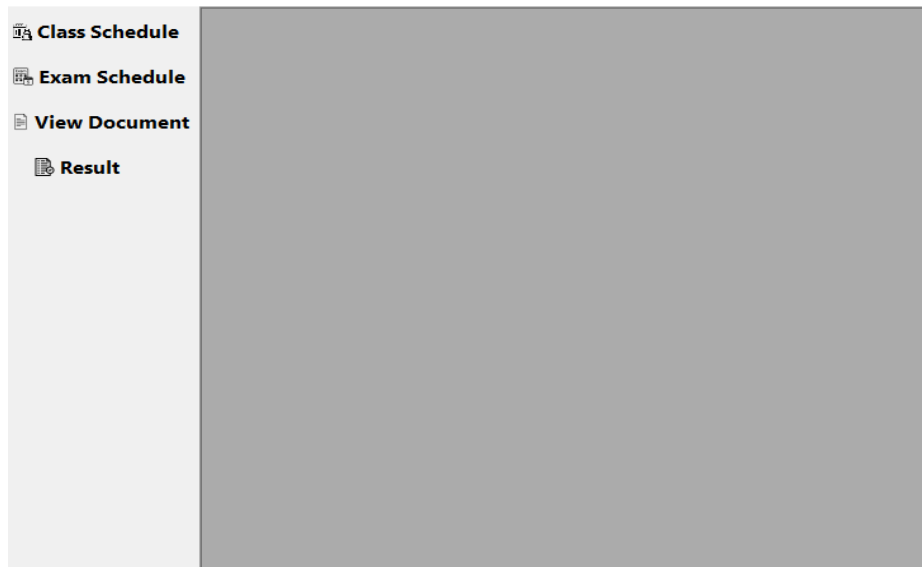
Admission

Registration

Student Result

Upload

Exam Schedule



Security Design

Authentication

- Username/password authentication
- Password policies (minimum length, complexity)
- Password encryption using industry standard algorithms
- Account lockout after multiple failed attempts

Authorization

- Role-based access control (Admin, Teacher, Student)
- Function-level permissions

- Data-level access restrictions
- Session management and timeout

Data Protection

- Encryption of sensitive data in database
- Secure connection to database
- Protection against SQL injection
- Input validation and sanitization

Audit Trails

- Logging of user actions
- System access logs
- Data modification logs
- Security event logs

Error Handling

Error Categories

- User input errors
- Business logic errors
- Database errors
- System errors

Error Handling Strategy

- User-friendly error messages
- Detailed logging for debugging
- Graceful degradation
- Exception handling and recovery

Error Logging

- Log file structure and location
- Error severity levels
- Log rotation strategy
- Critical error notifications

Performance Considerations

Performance Requirements

- Support for 500+ students and 50+ teachers
- Response time under 2 seconds for standard operations
- Efficient database queries
- Minimal memory footprint

Optimization Techniques

- Database indexing
- Query optimization

- Caching frequently accessed data
- Lazy loading for large datasets
- Pagination for long lists

Scalability Approach

- Modular design for easy extension
- Separable components
- Configurable parameters

Testing Strategy

Unit Testing

- Test individual components and functions
- Mock database connections
- Assert expected behaviors
- Cover edge cases and error conditions

Integration Testing

- Test component interactions
- Validate data flow between layers
- Ensure proper module communication

System Testing

- End-to-end testing of complete workflows

- Performance testing under expected load
- Security testing (penetration testing)

User Acceptance Testing

- Real-world usage scenarios
- User feedback collection
- Usability assessment

Deployment Strategy

Deployment Architecture

- Windows desktop deployment
- Central database server
- Local client installations

Installation Procedure

1. Database setup and configuration
2. Application installation
3. Connection configuration
4. Initial data setup

System Requirements

- Windows 10 or higher
- .NET Framework 4.7.2 or higher

- MS SQL Server 2019 or higher
- Minimum 4GB RAM
- 500GB storage space

Backup and Recovery

- Regular database backups
- Application configuration backups
- Recovery procedures documentation

Maintenance and Support

Maintenance Plan

- Regular updates and patches
- Bug fix release schedule
- Feature enhancement process
- Version control and release management
-

Appendices

Glossary

Administrator: School staff with full system access

CRUD: Create, Read, Update, Delete - basic database operations

Dashboard: Main page showing overview information

Module: Functional section of the application

Reference Documents

- System Requirements Specification
- UI Style Guide
- Database Schema Documentation
- Test Plan